

Project 2: Test Case Design & Execution for a Simple Web Application

2. Objective

- To design and manually execute test cases for an e-commerce web application and identify defects by testing positive, negative, and edge scenarios.

3. Key Features to Test

- User Registration
- User Login
- Product Search
- Product Details
- Add to Cart
- Cart Update/Remove
- Checkout
- Logout

4. Test Cases:1

- **TCID:**TCO1
- **Test Objective:** Verify registration with valid data
- **Preconditions:** Registration page is open
- **Steps:** Enter valid name, email, password and click Register

- **Expected Result:** Account should be created successfully
- **Actual Result:** Account created successfully
- **Status:** Pass

➤ **Test cases:2**

- **TCID:**TC02
- **Test Objective:** Verify registration with invalid email
- **Preconditions:** Registration page is open
- **Steps:**Enter invalid email and valid other details
- **Expected Result:**Error message should be displayed
- **Actual Result:** Error message displayed
- **Status:** Pass

➤ **Test cases:3**

- **TCID:**TC03
- **Test Objective:** Verify registration with empty fields
- **Preconditions:** Registration page is open
- **Steps:**Leave all fields blank and click Register
- **Expected Result:**Required field messages should appear
- **Actual Result :**Required messages displayed
- **Status:**Pass

➤ **Test cases:4**

- **TCID:**TC04
- **Test Objective:**Verify login with valid credentials
- **Preconditions:**User account exists
- **Steps:**Enter valid email and password
- **Expected Result:**User should login successfully
- **Actual Result :**Login successful
- **Status:**Pass

➤ **Test cases:5**

- **TCID:**TC05
- **Test Objective:**Verify login with wrong password
- **Preconditions:**User account exists
- **Steps:**Enter valid email and wrong password
- **Expected Result:**Error message should appear
- **Actual Result:**Error message displayed
- **Status:**Pass

➤ **Test cases:6**

- **TCID:**TC06
- **Test Objective:**Verify login with blank fields
- **Preconditions:**Login page is open
- **Steps:**Leave email and password blank
- **Expected Result:**Validation message should appear
- **Actual Result:**Validation message displayed
- **Status:**Pass

➤ **Test cases:7**

- **TCID:** TC07
- **Test Objective:** Verify product search with valid product
- **Preconditions:** Website is open
- **Steps:** Enter product name in search box
- **Expected Result:** Matching products should be displayed
- **Actual Result:** Matching products displayed
- **Status:** Pass

➤ **Test cases:8**

- **TCID:** TC08
- **Test Objective:** Verify search with unavailable product
- **Preconditions:** Website is open
- **Steps:** Search for a non-existing product
- **Expected Result:** "No products found" should appear
- **Actual Result:** No products found displayed
- **Status:** Pass

➤ **Test cases:9**

- **TCID:** TC09
- **Test Objective:** Verify adding product to cart
- **Preconditions:** Product is available
- **Steps:** Open product and click Add to Cart
- **Expected Result:** Product should be added to cart
- **Actual Result:** Product added

- **Status:**Pass

➤**Test cases:10**

- **TCID:**TC10
- **Test Objective:**Verify removing product from cart
- **Preconditions:**Product is in cart
- **Steps:**Open cart and click Remove
- **Expected Result:**Product should be removed
- **Actual Result:**Product removed:
- **Status:**Pass

➤**Test cases:11**

- **TCID:**TC11
- **Test Objective:**Verify quantity update
- **Preconditions:**Product is in cart
- **Steps:**Increase product quantity
- **Expected Result:**Cart quantity and price should update
- **Actual Result:**Quantity updated
- **Status:**Pass

➤**Test cases:12**

- **TCID:**TC12
- **Test Objective:**Verify checkout with valid details
- **Preconditions:**Product is in cart and user logged in
- **Steps:**Enter valid address and payment details
- **Expected Result :**Order should be placed successfully

- **Status:**Pass

➤ **Test cases:13**

- **TCID:**TC13
- **Test Objective:**Verify checkout with missing address
- **Preconditions:**Product is in cart
- **Steps:**Leave address blank and continue
- **Expected Result:**Address validation should appear
- **Actual Result:**Validation displayed
- **Status:**Pass

➤ **Test cases:14**

- **TCID:**TC14
- **Test Objective :**Verify logout
- **Preconditions:**User is logged in
- **Steps:**Click Logout
- **Expected Result:**User should be logged out
- **Actual Result:**User logged out
- **Status:**Pass

➤ **Test cases:15**

- **TCID:**TC15
- **Test Objective:**Verify maximum/edge quantity
- **Preconditions:**Product is in cart
- **Steps:**Enter a very large quantity

- **Expected Result** :Application should show proper validation
- **Actual Result**:Application accepts invalid quantity
- **Status**:Fail

5. Positive, Negative and Edge Scenarios

➤ **Positive Testing:**

- Testing the application with valid input.

Examples:

- Valid email and password
- Valid product search
- Adding an available product to cart
- Successful checkout

➤ **Negative Testing**

- Testing the application with invalid input.

Examples:

- Wrong password
- Invalid email
- Empty required fields
- Missing address during checkout

➤ **Edge Testing**

- Testing the application with boundary or unusual values.

Examples:

- Very large product quantity
- Very long username
- Minimum/maximum password length
- Very long search text

6. Bug Report

➤ **Bug ID:** BUG-001

➤ **Summary:** Application accepts an invalidly large product quantity.

➤ **Description:**

- The shopping cart allows the user to enter an extremely large quantity without displaying a validation message or restricting the quantity.

➤ **Steps to Reproduce:**

- Open the e-commerce website.
- Login with a valid account.
- Add any product to the cart.
- Open the cart.
- Enter a very large quantity, such as 999999.
- Click Update Cart.

➤ **Expected Result:**

- The application should restrict the maximum quantity or display an appropriate validation message.

➤ **Actual Result:**

- The application accepts the extremely large quantity without proper validation.

➤ **Severity:** Medium

➤ **Priority:** Medium